Homework 2

Problem 1

(a) For g(x) = (a(x), b(x)), take

$$\mu(x) = a(x), \qquad \sigma(x) = \log(1 + e^{b(x)}) > 0.$$

The mean is unrestricted. I chose softplus because it is positive and grows approximately linearly for large positive b. The exponential link is valid but grows exponentially. ReLU can output zero, giving a degenerate Gaussian. The variance is $\sigma(x)^2$.

(b) The Gaussian density is

$$p(y \mid x) = \frac{1}{\sqrt{2\pi}\,\sigma(x)} \exp\left[-\frac{(y-\mu(x))^2}{2\sigma(x)^2}\right].$$

Taking minus the log-density and averaging gives

$$L = \frac{1}{n} \sum_{i=1}^{n} \left[\log \sigma_i + \frac{(y_i - \mu_i)^2}{2\sigma_i^2} + \frac{1}{2}\log(2\pi)\right].$$

(c) [Code on GitHub] With seed 42, I split the data 70%/15%/15% into 366,487 training, 78,533 validation, and 78,534 test rows. The MLP is 7 → 64 → 64 → 2, with ReLU hidden layers. The seven inputs are already standardized. For the target, c = 10.416308 and s = 7.755263 are the training mean and standard deviation:

$$z = (y - c)/s, \qquad \mu_y = c + s\mu_z, \qquad \sigma_y = s\sigma_z.$$

I used Adam (learning rate 0.001), batch size 1024, at most 100 epochs, and patience 10 on validation NLL. Epoch 48 was selected. The baseline is the constant Gaussian $N(c, s^2)$. NLL in minutes equals standardized-target NLL + log s.

Model	Validation NLL	Validation RMSE (min)
Constant Gaussian	3.466067	7.745618
MLP Gaussian	2.630229	3.904418

(d) [Code on GitHub] Since (Y − μ(x))/σ(x) is conditionally standard normal,

$$p(x) = P(Y < 45 \mid X = x) = \Phi\left(\frac{45 - \mu(x)}{\sigma(x)}\right).$$

Here Φ is the standard normal CDF; μ and σ are in minutes.

(e) [Code on GitHub] On the test set, use

$$t_i = \mathbf{1}\{y_i < 45\}, \qquad \text{Brier} = \frac{1}{m} \sum_{i=1}^{m} (p_i - t_i)^2.$$

$$\text{Accuracy} = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}\{\mathbf{1}\{p_i \geq 0.5\} = t_i\}.$$

Model	NLL	MSE (min²)	RMSE (min)	Brier	Accuracy
Constant Gaussian	3.467520	60.169377	7.756892	0.004457	0.995543
MLP Gaussian	2.756783	16.163363	4.020369	0.002925	0.996677

The MLP improves all five measures. Since 99.55% of test trips are under 45 minutes, the Brier score is more informative than accuracy alone.

Problem 2

(a) [Code on GitHub] StockCode has 3,665 categories and CustomerID has 4,339. Sparse one-hot encoding is feasible, but a dense float32 matrix would use about 12.74 GB. I used the same seed-42 split for all models: 278,546 training, 59,689 validation, and 59,689 test rows.

Quantity has median 6 and maximum 80,995. I kept all observations and used squared loss on log(1 + Q) to reduce the influence of extreme values:

$$L=\frac{1}{n}\sum_{i=1}^{n}(f(x_i)-\log(1+q_i))^2, \qquad \hat{q}(x)=e^{f(x)}-1.$$

The back-transform estimates exp(E[log(1 + Q) | X]) - 1.

(b) [Code on GitHub] I used five-fold target encoding (seed 42). For feature j, define

$$t_j(c;S)=\frac{\sum_{i\in S} q_i \mathbf{1}\{x_{ij}=c\}}{\sum_{i\in S}\mathbf{1}\{x_{ij}=c\}}.$$

For a training row, S contains the other four folds. For validation and test, S is the full training set. Unseen categories receive the mean quantity in S. I logged the two features and standardized them using their OOF training mean and standard deviation, then fitted a 2 → 64 → 32 → 1 ReLU MLP.

(c) [Code on GitHub] Each categorical feature has a separate 16-dimensional embedding. Their concatenation feeds a 32 → 64 → 32 → 1 ReLU MLP. Unseen categories use a zero embedding.

(d) [Code on GitHub] Hybrid concatenates the two encoded features and both embeddings, giving a 34 → 64 → 32 → 1 ReLU MLP.

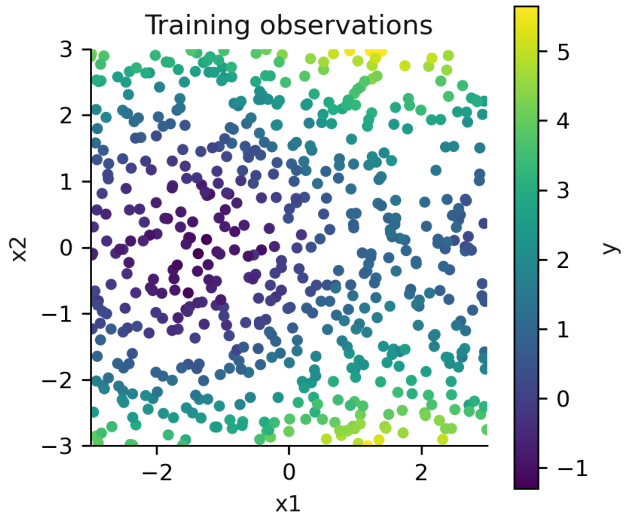
All three use Adam (learning rate 0.001), batch size 1024, and at most 50 epochs. Early stopping restores the best validation log1p-MSE weights, with patience 5.

Model	Epoch	Validation log1p-MSE	Test log1p-MSE	Test MAE	Test RMSE
Target encoding	22	0.387451	0.388380	6.927856	32.806744
Embedding	29	0.292530	0.291603	6.223117	31.490660
Hybrid	22	0.295415	0.294542	6.231761	31.608148

(e) [Code on GitHub] Target encoding retains only a marginal mean, which large purchases can distort. Embeddings learn several properties per category and allow stock-customer interactions. Rare categories provide few observations, so their embeddings can overfit. Both embedding models improve on target encoding. Embedding has the lowest test log1p-MSE, MAE and RMSE. Adding target means did not improve the embedding model here. The means may duplicate information in the embeddings or add noise from extreme purchases. This ranking need not hold for other data splits.

Problem 3

(a) [Code on GitHub] I kept the supplied 600/200/200 splits. The response rises toward large |x2| and has a low region near (−1, 0).



(b) [Code on GitHub] With seed 42, h = 256 and σ = 1, I used

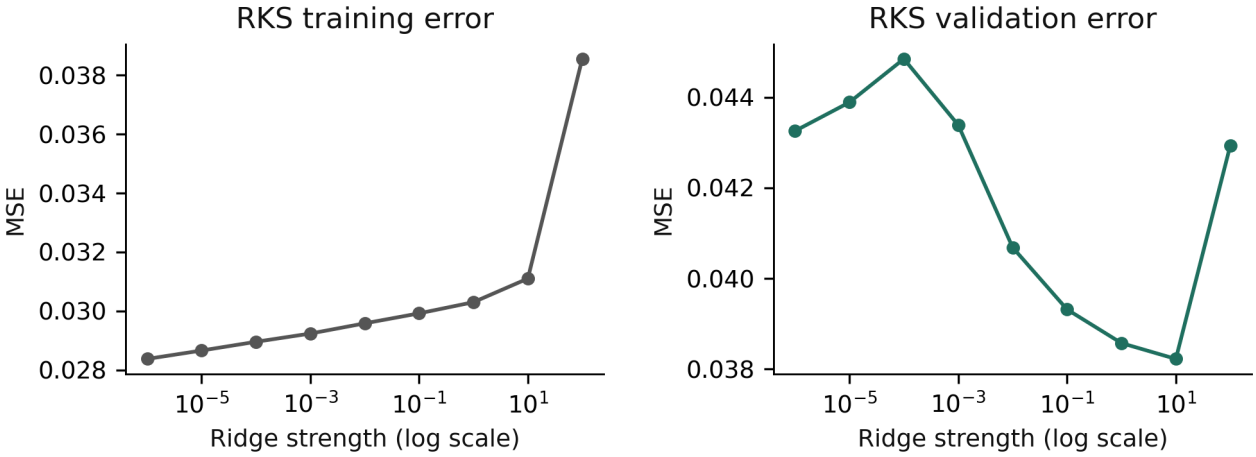
$$z(x) = (\sin(w_1^T x + b_1), \cos(w_1^T x + b_1), \ldots, \sin(w_h^T x + b_h), \cos(w_h^T x + b_h))^T.$$

$$w_j \sim N_2(0, I_2/\sigma^2), \qquad b_j \sim U[0, 2\pi], \qquad D = 2h = 512.$$

The same random features are used for all fits and splits. Ridge regression minimizes

$$\min_{\beta, c} \sum_{i \in \text{train}} (y_i - c - z(x_i)^T \beta)^2 + \lambda \|\beta\|_2^2.$$

The intercept is unpenalized. I scanned λ = 10^{-6}, 10^{-5}, ..., 10^{2}, keeping the other settings fixed.



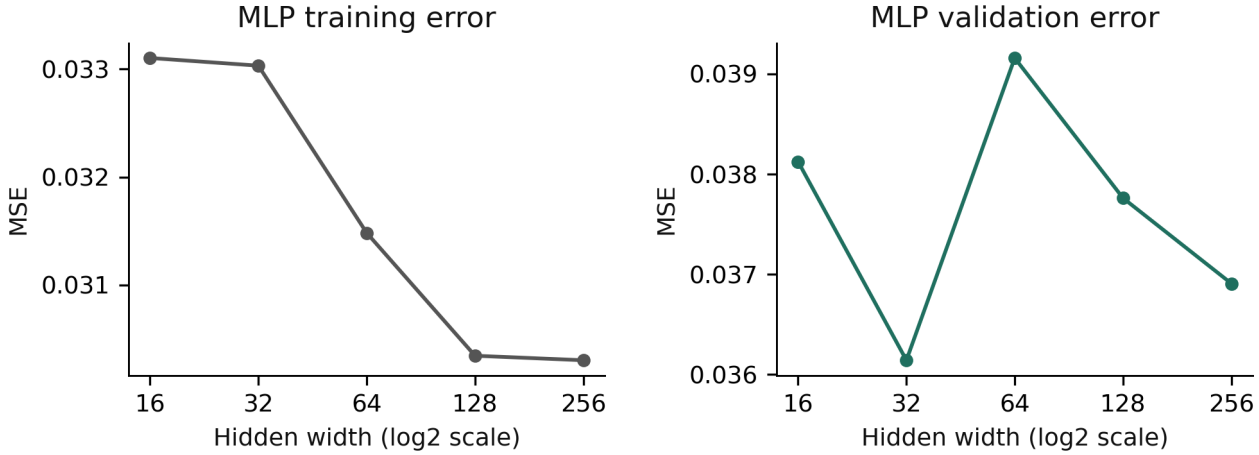
Validation selects λ = 10, with MSE 0.038224. Small λ values give lower training error but higher validation error, suggesting overfitting. Validation MSE falls from λ = 10^{-4} to 10; at λ = 100, both errors increase.

Problem 3 (continued)

(c) [Code on GitHub] The single-hidden-layer MLP is

$$f(x) = v^T \tanh(Wx + b) + c.$$

I scanned widths 16, 32, 64, 128, and 256. Each fit used seed 42, full-batch Adam with learning rate 0.005, and 3000 epochs.

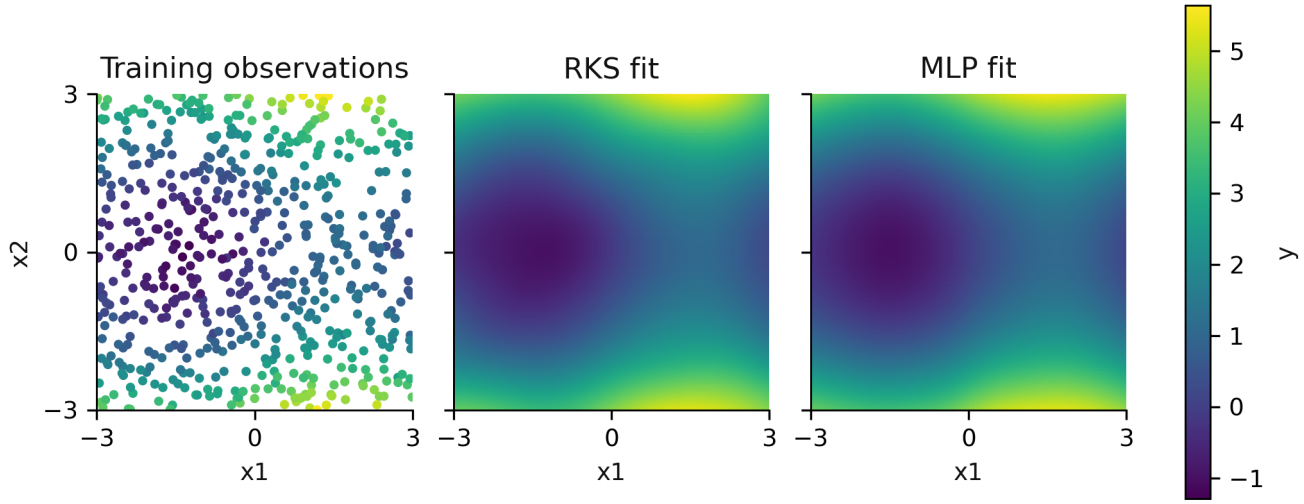


Training MSE falls as width increases, but validation MSE is nonmonotonic. Width 32 has the lowest validation MSE; wider networks fit the training data better without improving validation performance here.

(d) [Code on GitHub] I refitted both selected models on train only, with the same seed and settings.

Refitted model	Train MSE	Validation MSE
RKS	0.031101	0.038224
MLP	0.033032	0.036143

The MLP has slightly lower validation MSE. Its learned features can adapt to this response, while the RKS features are fixed.



Both fits capture the rise at large |x2| and the trough near (−1, 0) in (a), smoothing nearby observations. The color scale matches (a).

Both CSVs (`rks_test_predictions.csv`, `mlp_test_predictions.csv`) have 200 rows in test order, no header and no index. Test labels are unavailable, so test MSE cannot be computed.